

NIRAV SURABHI

Software + mechanical engineering, from Civic aerodynamics to bias-reducing news apps.

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EDUCATION

Purdue University B.S. Mechanical Engineering	FALL 2026 · INCOMING
Independence High School Loudoun County, VA · Senior	GRAD. JUN 2026
Coursework: AP Computer Science A, AP Physics C, Multivariable Calculus.	
Central Virginia Governor's School Science & Technology · Mathematics scholar	JUN-JUL 2024

MECHANICAL / HARDWARE

Passive Vortex Propulsion	2024-2025 · 1ST AUTHOR
Started as an experiment to make my Honda Civic more efficient; built a wind tunnel from scratch, soldered & calibrated load cells with an Arduino, and ran CFD in Fusion 360. First demonstration of passive vortex propulsion in air — published first-author paper; AIAA award & NASA Goddard invitation.	
Arduino · C++ · load cells · Fusion 360 · CFD · Python	
FTC Robotics — Absolute Zero (12096)	2022-2026 · CAPTAIN
Four years as captain & lead programmer; CADded and built every part in Fusion 360. Wrote autonomous & tele-op in Java — PID loops, sensor fusion, odometry. Reached the Chesapeake championship.	

EV & Salinity Sensor	SCIENCE OLYMPIAD
Autonomous EV that accelerates then brakes to a target distance (Arduino, goBILDA motor, quadrature encoder, PID); Arduino salt-water salinity sensor reading conductivity.	

SKILLS

LANGUAGES	Java, Python, JavaScript, TypeScript, HTML, CSS, Git
AI / ML	Prompt engineering, LLMs, CNNs, computer vision, agent architecture
CLOUD	Fly.io, Vercel, Supabase, Fusion 360
HARDWARE	Arduino, load cells, soldering, multimeter, PID control

HONORS & MUSIC

- AIAA Aerospace Award (RSEF special honor → NASA Goddard invitation)
- Captain, Science Olympiad
- Presidential Volunteer Service Award (Silver)
- Central Virginia Governor's School for Science & Technology — Mathematics, 2024
- 1st place, Agaudo guitar competition · All-Virginia Guitar Ensemble

SOFTWARE — FOUNDER PROJECTS

Chad vs. Chud chadvschud.com	2026-NOW · CO-FOUNDER/CEO
Real-time 1v1 webcam face-rating battle scored by an AI. Built the Next.js client, a Socket.IO/WebRTC matchmaking service, and a Python/FastAPI inference service (ResNet-18 + ArcFace + LLM scorer). 10,000+ users, 100+ concurrent.	
Next.js · TypeScript · FastAPI · WebRTC · PyTorch · Supabase · Fly.io	
b2alpha.io b2alpha.io	2026-NOW · CO-FOUNDER/CEO
Open agent-to-agent protocol — secure messaging + DNS integration; prompt engineering for structured, parseable output. 2,000+ business users.	
TypeScript · Node.js · OpenAI API · Vercel · Supabase · DNS	
PredictionView predictionview.app	2026-NOW · CO-FOUNDER/CEO
Full-stack Kalshi prediction-market trading hub — charts, an OpenAI analyst with web research, alerts, watchlists, a journal/backtester. Own the Express + Prisma/Postgres backend (127 routes, cron jobs) and Vitest tests.	
React · TypeScript · Vite · Express · Prisma · PostgreSQL · OpenAI	
Khali khali.info	2025-2026 · CO-FOUNDER/CEO
TikTok-style swipe-through news startup built to make short-form scrolling useful: stories combine opposing sources, present one claim at a time, and include an OpenAI "ask this story" assistant. Native SwiftUI iOS + Expo/RN app on a Node + Python backend.	
Swift/SwiftUI · React Native · Expo · Node.js · Python · OpenAI · Supabase	
EXPERIENCE	
AI Engineer & SWE Intern Ember Learning (formerly DeAP Learning Labs)	FEB 2021-NOW
Wrote the LLM prompts behind AP-tutor chatbots; built automated AI grading for AP DBQs/FRQs against the official rubric; indexed lecture transcripts in a vector DB for timestamp search; led the company website. Team landed a Microsoft & OpenAI grant; \$1M+ valuation.	
CAD Intern I3 Simulations	JUN 2025-MAY 2026
3D models and haptic-feedback prototypes for medical-training simulators.	
Bike Mechanic The Bike Lane · intern → part-time	JUN 2021-AUG 2025
Built bikes from the frame up; ran final safety checks.	